

Systems Thinking in Large Capital Project Management

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Executive Summary

Large capital projects fail not because individual components are poorly engineered but because the couplings between components — regulatory, operational, financial, social — are modelled too late, if at all. Analysis of fifteen landmark cases spanning nuclear power stations, rail programmes, carbon capture schemes, hospital IT, and mining projects reveals five recurring systemic blindspots that precede virtually every major overrun and cancellation.

This paper introduces *systems-thinking*, a Claude Code skill that operationalises fifteen research-derived heuristics into a structured audit framework. Applied before a Final Investment Decision or stage gate, the skill produces a board-ready report covering all fifteen SYS rules, a maturity score on a one-to-five scale, a ranked gap list using a standardised omission vocabulary, and a recommended tool stack for closing identified gaps.

Evaluation against three test scenarios — an offshore wind and green hydrogen project, a hospital electronic health record transformation, and a first-of-a-kind lithium hydroxide refinery — shows that outputs produced with the skill pass 100 percent of structured assertions, against 41 percent without it. The skill adds comparability and completeness: both configurations correctly identify domain-specific risks, but only the skill-guided outputs apply the same fifteen-rule framework and standardised vocabulary consistently, making cross-project comparison tractable.

The paper describes the rationale, the architecture, the evaluation, and the usage guide for practitioners deploying the skill in project governance.

1. The Systemic Failure Pattern

1.1 Beyond Budget and Schedule

The most widely cited measure of megaproject underperformance is cost overrun. Flyvbjerg's cross-sectoral database of 16,000 projects established that 91.5 percent exceed their original budget or schedule targets, and only 0.5 percent deliver all promised benefits on time and on budget.¹ McKinsey's analysis of over 300 projects exceeding USD 1 billion found average cost overruns of 80 percent and schedule delays of 50 percent.²

These figures, while sobering, describe symptoms rather than causes. The deeper pattern — visible in the best-documented failure cases — is that projects are scoped as asset construction programmes when their success depends on systems that extend far beyond the built object. A nuclear power station requires not just reactor installation but integrated quality assurance, sub-contractor oversight, and regulatory documentation as an indivisible whole. A carbon capture project requires not just the capture unit but a financeable full-chain economics spanning carbon pricing, transport tariffs, and storage contracts. A hospital IT programme requires not just software deployment but clinical workflow integration, trust autonomy, and local change capability.

When the scope definition omits these dependencies, projects build part of their failure in from the start.

1.2 Fifteen Cases, Five Blindspots

Analysis of fifteen landmark projects — selected because authoritative public post-mortems exist in sufficient detail to support root-cause attribution — reveals five recurring systemic blindspots.³

Blindspot 1: Legal-regulatory non-integration. The Longannet CCS project in the United Kingdom was cancelled when it became clear that funding caps, the carbon price floor, and the contract architecture formed an incompatible economic system that could not be made financeable. The regulatory environment was not a boundary condition to be managed — it was a structural component of the project’s value thesis, and it was modelled too late. Pascua-Lama in Chile and Argentina followed the same logic: environmental regulation and indigenous rights were treated as downstream permitting tasks until court decisions made the project unbuildable.⁴

Blindspot 2: Build-before-design-freeze. Olkiluoto 3, Flamanville 3, Vogtle, and the Sydney Opera House are four variants of one pattern: political or commercial pressure for a visible start meets insufficiently mature design, incompletely validated manufacturing paths, or immature supply chains. The result is not merely rework; it is a qualitatively different project with newly created dependencies. At Olkiluoto 3, the design-fabrication-quality assurance loop was not treated as an integrated system — STUK investigations found that subcontractor control and regulatory documentation were separated from technical design in a way that made early cost and schedule estimates unrecoverable.⁵

Blindspot 3: Late integration economics. Crossrail and the NHS National Programme for IT both featured many subsystems that appeared “green” in isolation while the overall system remained non-operational. For Crossrail, civil works were mastered years ahead of signalling, station systems, safety assurance, and the thousands of physical-digital asset interactions that defined operational readiness. The NAO investigation found that the integration and testing phase had been systematically planned too small and started too late.⁶ NPfIT collapsed for the equivalent reason in software: centralised architecture, heterogeneous legacy trust systems, and clinical workflow diversity were not integrated ends of the same delivery equation.

Blindspot 4: Physical-social coupling. The Hallandsås Tunnel in Sweden took 23 years and cost roughly ten times its original budget. The root cause was not geological surprise alone — it was that geology, groundwater chemistry, environmental law, and local legitimacy were treated as separate streams rather than a coupled physical-legal-social system. Once toxic injection agents entered the groundwater, the project became politically redefinable in ways that no schedule or cost model had anticipated. Pascua-Lama replicated this pattern: the project’s physical footprint and its legal footprint diverged irreconcilably.⁷

Blindspot 5: Governance fragmentation. The Scottish Parliament Building, the Big Dig in Boston, and Crossrail each suffered from accountability for scope, cost, quality, and interfaces being distributed across organisations, contracts, and political levels in ways that made systemic risk invisible at any single node. The Holyrood Audit Committee found that brief, design, and procurement stabilised too late precisely because no single governance structure had authority across the full system boundary.⁸

These five blindspots are not coincident failures of project management rigour. They are structurally predictable consequences of defining complex sociotechnical systems as if they were asset

construction projects.

1.3 The Upshot for Project Definition

The implication is straightforward but uncomfortable: a project definition that describes the physical asset without modelling the full system — feedstock, grids, permits, tax and levy treatment, operating model, supply-chain readiness, social licence, and exit logic — is not a definition. It is an engineering brief for one component of a system whose other components have not been described.

The research is explicit on this point: **“The skill must never treat project definition as asset definition alone.”**⁹ This design principle drives every element of the *systems-thinking* skill.

2. The Case for a Structured Audit Skill

2.1 The Problem with Ad Hoc Review

Experienced project directors, supervisory board members, and independent technical advisers routinely identify systemic risks in project definitions. The challenge is that unstructured review is inconsistent, non-comparable, and dependent on the reviewer’s particular experience base. A reviewer whose background is energy markets may rigorously interrogate pricing assumptions while missing interface management gaps. One with infrastructure experience may flag integration risks while overlooking social licence dynamics.

The result is that the quality of a pre-FID systemic review correlates strongly with the identity of the reviewer, not the characteristics of the project. This makes governance a function of personnel rather than process — an arrangement that is difficult to audit, impossible to mandate, and unlikely to scale across a portfolio.

2.2 What a Skill Adds

A Claude Code skill packages structured analytical behaviour that is applied consistently regardless of context. The *systems-thinking* skill does not add domain knowledge that an experienced reviewer lacks — both the skill-guided and unskilled configurations of Claude correctly identify that a lithium hydroxide project’s business case depends on lithium prices, or that a Finnish FOAK refinery requires an environmental permit under the YVA procedure. The skill’s value is different: it guarantees that the same fifteen analytical tests are applied in the same sequence with the same vocabulary, every time, regardless of project sector.

This has three consequences. First, gaps that fall outside the reviewer’s habitual focus cannot be skipped — the framework forces completion across all fifteen rules. Second, findings are expressed in a standardised vocabulary, making cross-project comparison tractable in a way that prose summaries cannot be. Third, the output format is board-ready by design: a supervisory board member reviewing the report for a hydrogen project and the report for a hospital IT programme can compare maturity scores, flag counts, and omission vocabulary across both, without translating between different analysts’ frameworks.

2.3 Marketplace Skill Leverage

Three existing Claude Code marketplace skills are integrated as recommended follow-on tools, matching specific SYS rules to purpose-built capabilities:

Skill	SYS rule(s)	Use case
what-if-oracle	SYS-11	Scenario stress test: carbon price $\pm 50\%$, demand -20% , interest rates $+200\text{bp}$, permit delay $+24$ months
perplexity-search	SYS-03	Regulatory due diligence: jurisdiction-specific permit requirements, levy classifications, subsidy status
literature-review	SYS-07	Reference class forecasting: comparable projects, published cost-to-complete distributions, IPA benchmarks

The skill does not auto-invoke these tools; it flags their applicability in the recommended next actions section, preserving user control over which follow-on analyses to commission.

3. Architecture and Design Principles

3.1 Two-File Structure

The skill is implemented as two files:

```
~/.claude/skills/systems-thinking/  
+-- SKILL.md          -- workflow, output template, tone guidance  
+-- references/  
    +-- sys-rules.md   -- detailed criteria for all 15 SYS rules
```

The separation follows the Claude Code progressive disclosure principle: `SKILL.md` contains the audit workflow and exact output template (under 500 lines), while `sys-rules.md` is loaded on demand when applying the rules requires deeper criteria. This keeps the primary skill file within the context budget for all but the most resource-constrained deployments.

`SKILL.md` carries YAML frontmatter with a description that functions as the primary triggering mechanism. The description lists the specific phrase contexts that should activate the skill — “systems thinking audit”, “check for systemic risks”, “megaproject risk review”, “SYS audit”, “what could go wrong with this project”, as well as implicit contexts such as a project director sharing a project document before a board approval gate.

3.2 The Five-Step Audit Workflow

The skill applies a defined five-step process:

1. **Frame the project** — establish type, phase, sector, decision gate, and information provided. Identify which of the fifteen rules are most likely to surface material gaps given the project characteristics.
2. **Apply all fifteen SYS rules** — rate each rule PRESENT, PARTIAL, or ABSENT based on the evidence provided. For PARTIAL and ABSENT ratings, write a finding that names the specific gap and its systemic consequence.
3. **Score the maturity level** — assign an overall maturity score on the 1–5 scale (see Section 3.5), calibrated to the distribution of PRESENT, PARTIAL, and ABSENT ratings and the severity of the highest-priority gaps.
4. **Rank critical gaps** — select the three to five gaps posing the greatest systemic risk, ordered by the combination of impact severity and reversibility. A gap that is remediable at pre-FID cost but catastrophic if discovered post-FID ranks above a gap that is material but discoverable and correctable during execution.
5. **Recommend next actions** — map gaps to tools from the minimal stack (see Section 3.4) and to the three integrated marketplace skills, producing a prioritised action table with owner, output, and timeframe.

3.3 The Fifteen SYS Rules

The fifteen rules are derived directly from the failure patterns in the fifteen case studies. Each rule addresses a recurring failure class, states its rationale in one sentence, and specifies the PRESENT, PARTIAL, and ABSENT criteria that allow consistent rating across reviewers and projects. Appendix A contains the full table; the rules are:

Rule	Short title	Core failure addressed
SYS-01	Full value chain	Missing upstream/downstream dependencies
SYS-02	Second and third-order effects	Indirect systemic consequences
SYS-03	Regulatory map before FID	Late legal and levy discoveries
SYS-04	Design-maturity threshold	Build-before-design-freeze
SYS-05	Interface ownership and evidence	Unmanaged system interfaces
SYS-06	Risk-adjusted estimate	False certainty in point budgets
SYS-07	Outside view first	Systematic optimism bias
SYS-08	Integration as its own megaproject	Late and under-resourced integration
SYS-09	Supply-chain and workforce readiness	FOAK supply-chain brittleness
SYS-10	Stakeholders as system components	Social licence as project risk
SYS-11	Business case stress test	Fragile economics under policy/price scenarios
SYS-12	Explicit kill criteria	Sunk-cost traps in prestige projects

Rule	Short title	Core failure addressed
SYS-13	Operations and maintenance in definition	ORAT and handover gaps
SYS-14	FOAK as learning programme	First-of-a-kind treated as serial production
SYS-15	Cluster and platform level	Sub-optimal standalone asset solutions

The rules are not a checklist to be completed mechanically. They constitute a system: SYS-04 (design maturity) and SYS-09 (supply-chain readiness) compound in FOAK programmes, as Olkiluoto 3 and Vogtle showed. SYS-03 (regulatory map) and SYS-11 (business case stress test) compound in market-dependent projects such as CCS or energy storage, where the regulatory framework directly determines the economics. Reviewers are instructed to identify and surface these compounding interactions, not just score each rule in isolation.

3.4 The Minimal Tool Stack

The research identifies a minimal stack adequate for early project definition review. The skill maps each stack element to its corresponding SYS rules:

Tool	SYS rules	What it delivers
Value-Chain Map	SYS-01, SYS-02	End-to-end dependency visibility
Regulatory Map	SYS-03	Permits, levies, liability, subsidies, tax treatment
Interface Register	SYS-05	Interface ownership and maturity evidence
ConOps / Operating Model	SYS-13	Day-in-the-life operational scenario
Scenario Stress Test	SYS-11	Business case at policy and price extremes
Monte Carlo Cost/Schedule	SYS-06	P50/P80/P90 risk-adjusted estimates
Reference Class Forecast	SYS-07	Outside view on cost and schedule
Supply-Chain Readiness Assessment	SYS-09	Top-10 supplier and craft labour gaps
Independent Red Team / Gate Challenge	SYS-12	External counter-review with kill criteria

3.5 Maturity Scoring

The maturity score aggregates the rule ratings into a single governance-facing signal:

Score	Description	Typical profile
1	Asset-centric only	Full chain absent; no regulatory map; interfaces undefined; integration not planned
2	Partial chain with major gaps	Chain sketched but incomplete; PARTIAL on most rules; at least one ABSENT on critical rules (SYS-03, SYS-06, SYS-07)
3	Structured approach, incomplete evidence	Framework present; key elements PARTIAL; integration and ORAT underweighted
4	Mature definition with minor gaps	Most rules PRESENT; remaining PARTIAL items are manageable; outside view applied
5	Fully integrated systems perspective	All rules PRESENT; chain, interfaces, regulation, operations, and exit logic complete

Most real projects reviewed to date score 2–3. A score of 4 or above at pre-FID is exceptional and merits verification that it reflects genuine evidence rather than optimism.

3.6 The Omission Vocabulary

Twelve standardised omission flags are embedded in the output template. Their purpose is to make gap identification directly comparable across projects and reviewers, replacing ad hoc language with a controlled vocabulary:

1. Full value chain not modelled
2. Regulatory due diligence missing
3. Interface owner unknown
4. Integration proof absent
5. Outside view missing
6. Kill criteria undefined
7. Design maturity not validated
8. Supply-chain readiness untested
9. Business case not stress-tested
10. FOAK not treated as learning programme
11. Stakeholder legitimacy not mapped
12. O&M not in scope of definition

Where a flag applies, the skill is instructed to use the exact phrase — not a paraphrase — in the Omission Flags section of the report. This enables automated screening of audit outputs across a project portfolio and makes it straightforward to track whether identified gaps are resolved at subsequent gate reviews.

4. Evaluation Approach and Results

4.1 Methodology

The skill was evaluated using the *skill-creator* framework, which runs parallel Claude Code agent instances — one with the skill loaded, one without — against the same test prompts, then grades outputs against structured assertions.

Three test scenarios were designed to cover different project types, decision gates, and audience profiles. For each scenario, both configurations produced a full audit report. The reports were graded against assertions verifiable from the text. Timing and token data were recorded at completion.

The evaluation measures what the skill adds relative to an unaided capable Claude model, not the absolute quality of either output.

4.2 Test Scenarios

Scenario 1: Offshore Wind and Green Hydrogen, Norwegian Continental Shelf (pre-FID)

A 500 MW offshore wind installation combined with a hydrogen electrolysis facility, targeting green hydrogen export via a proposed TSO pipeline network. The scenario provides a project brief including installed capacity targets and high-level CAPEX figures. The decision gate is FID; the audience is the project steering committee.

The systemic risks in this scenario include an unmodelled hydrogen export chain (SYS-01), early-stage TSO pipeline planning (SYS-03), and a business case that has not been stress-tested against carbon price and demand fluctuations (SYS-11).

Scenario 2: Hospital Electronic Health Record Transformation, Germany (supervisory board update)

A €340 million EHR transformation programme spanning eight German hospitals, with Phase 1 declared complete but Phase 2 — clinical integration and workflow adoption — not started. The scenario provides a programme update presented to the supervisory board.

The systemic risks in this scenario include the critical distinction between software installation and clinical integration (SYS-08), undefined criteria for continuation or termination given sunk costs (SYS-12), and an operating model that has not been validated in clinical workflow terms (SYS-13).

Scenario 3: FOAK Lithium Hydroxide Refinery, Finland (PMO stage-gate)

A €1.1 billion first-of-a-kind LiOH refinery project based on Australian spodumene feedstock, projecting an 18% IRR at prevailing lithium prices. The scenario provides a stage-gate pack for a PMO review. The project has not yet obtained Finnish environmental permits.

The systemic risks in this scenario include an unverified IRR in a commodity market that lost 80% of its value over 24 months (SYS-11), missing YVA environmental permit in Finland (SYS-03), FOAK technology risk treated as a standard EPC delivery problem (SYS-14), and a single-source spodumene feedstock from one Australian mine (SYS-09).

4.3 Assertions

Assertions were designed to test structural and vocabulary compliance, not domain knowledge. Domain-knowledge assertions — “does the output identify the hydrogen pipeline gap?” — were expected to pass in both configurations, because an unaided Claude model has adequate knowledge of hydrogen value chains, Finnish permitting law, and lithium market dynamics. The discriminating assertions test whether the skill-mandated framework is applied:

- **contains_maturity_score** — output contains an explicit X/5 score
- **covers_all_15_rules** — output contains SYS-01 through SYS-15 with ratings
- **uses_rating_vocabulary** — PRESENT / PARTIAL / ABSENT used throughout
- **uses_omission_vocabulary** — at least one exact omission flag from the twelve-item vocabulary
- **identifies_critical_gaps** — ranked gap section with at least three gaps
- **recommends_tool_stack** — specific named tools recommended in next actions

Domain-specific assertions varied by scenario: `flags_hydrogen_pipeline_gap`, `flags_integration_gap`, `flags_kill_criteria`, `flags_foak_risk`, `flags_permit_gap`, `flags_business_case_stress`.

4.4 Results

Configuration	Pass rate	Mean time	Mean tokens
With skill	100%	190.6 s	26,805
Without skill	41%	335.1 s	24,586
Delta	+59 pp	−144.5 s	+2,219

The 59-percentage-point pass rate delta is driven entirely by the structural and vocabulary assertions. Without the skill:

- Scenario 1 invented an eight-dimensional framework; no SYS-01 to SYS-15 labels appeared; omission vocabulary was absent; the tool-stack recommendation was generic
- Scenario 2 used eight “systems thinking” dimensions (stocks, flows, feedback loops); integration was discussed but the installed-vs-integrated distinction was not named with the skill’s vocabulary; kill criteria were addressed only in general terms
- Scenario 3 used ten categorical dimensions with CRITICAL/HIGH/MEDIUM severity ratings; FOAK risk was discussed without SYS-14; no exact omission flags appeared

In all three scenarios, the without-skill output correctly identified the material domain risks. The score penalty was structural, not substantive. This is precisely the value proposition: the skill does not supply knowledge that an experienced reviewer lacks. It supplies the framework that makes outputs comparable, complete, and auditable.

The with-skill outputs were also faster by an average of 144.5 seconds. The structured fifteen-rule framework reduced decision overhead: instead of constructing an analytical framework de novo, the agent applied a defined one. The small token premium (+2,219) reflects the additional structure loaded from the skill file.

4.5 Strongly and Weakly Discriminating Assertions

A post-hoc analysis of assertion behaviour identified two categories:

Strongly discriminating (skill 100%, baseline 0%): All structural and vocabulary assertions — `covers_all_15_rules`, `uses_rating_vocabulary`, `uses_omission_vocabulary`, `recommends_tool_stack`. These assertions fail in every without-skill run because the unaided model invents a different framework each time.

Non-discriminating (both pass): Domain-knowledge assertions — whether the output identifies the hydrogen pipeline gap, the Finnish permit requirement, the 18% IRR stress-test problem, the spodumene supply-chain risk. Both configurations pass these in all three scenarios. The conclusion is confirmed: the skill does not supply domain knowledge; it supplies structural discipline.

5. Usage Guide

5.1 Installing the Skill

The skill is distributed as a `.skill` package file. Installation requires one command:

```
# From the skill-creator directory
python3 -m scripts.package_skill /path/to/systems-thinking

# To install, place the unpacked skill in the skills directory
cp -r systems-thinking ~/.claude/skills/
```

After installation, Claude Code discovers the skill at startup. No configuration is required.

5.2 Invoking the Skill

The skill triggers on any of the following phrase patterns:

- "audit this project for systemic risks"
- "systems thinking audit" Or "SYS audit"
- "check this project definition for blind spots"
- "megaproject risk review"
- "what could go wrong with this project?"
- "review this FID pack / stage-gate document / project update"
- Sharing a project document and asking for a governance or risk review before a board, steering committee, or stage-gate

The skill also triggers when context suggests a project director, PMO lead, or supervisory board member is stress-testing a project before a major decision. No exact trigger phrase is required; the description is written to trigger on intent, not keywords.

5.3 What Input to Provide

The skill can work from any project documentation available. The richer the input, the more specific the findings. Useful inputs include:

- **Project brief or definition** — scope, CAPEX, timeline, technology, location, decision gate
- **Business case or investment paper** — financial model, IRR assumptions, market context
- **Programme update or status report** — progress against plan, issues log, next milestones

- **FID or stage-gate pack** — technical feasibility, regulatory status, contracting strategy, risk register

Where specific information is unavailable, the skill rates the relevant rule ABSENT and flags the omission. Absence of evidence is itself a finding: if the regulatory map has not been produced, the project does not have a regulatory map, and SYS-03 is ABSENT regardless of whether the regulatory risk is subsequently manageable.

5.4 Reading the Output

The audit report has a defined structure:

```
## Project Overview
[Project name, type, phase, decision gate, audience]

## Overall Maturity Score: X/5
[One-sentence characterisation of the maturity level]

## SYS Rule Assessment

| Rule | Rating | Finding |
|-----|-----|-----|
| SYS-01 | PRESENT / PARTIAL / ABSENT | [Specific finding or confirmation] |
...

## Critical Gaps (ranked)
[Top 3-5 gaps, ordered by systemic impact and reversibility, with brief rationale]

## Recommended Next Actions
[Table of gap → tool → owner → output → timeframe]

## Omission Flags
[Exact flags from the twelve-item vocabulary that apply to this project]

## Board Guidance Note
[Optional: plain-language summary for a non-technical board audience]
```

The overall maturity score is the governance-facing signal: it summarises the distribution of PRESENT, PARTIAL, and ABSENT ratings into a single comparable metric. The critical gaps section is the action-facing signal: it tells the project team where to direct effort before the next decision gate. The omission flags are the audit-facing signal: they produce a machine-readable list that can be tracked across gate reviews to verify closure.

5.5 Follow-On Tool Stack

After an initial audit, three follow-on workflows are most commonly indicated:

Scenario stress test (when SYS-11 or SYS-12 is ABSENT or PARTIAL): The what-if-oracle skill takes the business case as input and runs structured sensitivity analysis: carbon price at +50% and –50%, demand at –20%, interest rates at +200 basis points, permit delay at +24 months. The

output identifies which combinations breach the kill threshold, which provides the board with a concrete answer to “how fragile is the economics?”

Regulatory due diligence (when SYS-03 is ABSENT or PARTIAL): The *perplexity-search* skill retrieves current permit requirements, levy classifications, and subsidy status for the project’s jurisdiction. This is particularly valuable for cross-border projects and sectors with rapidly changing regulatory frameworks (CCS transport and storage, offshore hydrogen, critical minerals processing).

Reference class forecasting (when SYS-07 is ABSENT or PARTIAL): The *literature-review* skill retrieves published cost and schedule outcomes for comparable projects, enabling an outside-view correction to the current estimate. It covers GAO’s cost estimating benchmarks, IPA’s sector databases, NAO’s over-optimism analysis, and DOE’s FOAK lessons-learned series.

These three tools address the three assertions most commonly found ABSENT or PARTIAL in initial audits. Applying them in sequence converts a “maturity 2” project definition into a defensible stage-gate submission.

Appendix A — The Fifteen SYS Rules

Rule ID	Short Rule	Rationale	PRESENT criteria	PARTIAL criteria	ABSENT criteria
SYS-01	Full value chain mapped	Many failures occur outside the built object	Chain from feedstock/input to operations, maintenance, and exit is documented and owned	Chain sketched but missing upstream, downstream, or exit logic	Only the physical asset is described
SYS-02	Second and third-order effects	Systemic damage arrives indirectly	Indirect effects on grid, operations, tax, permitting, neighbours, and maintenance have been assessed	Some secondary effects identified; no systematic coverage	Effects assessed only at first order
SYS-03	Regulatory map before FID	Late legal and levy findings destroy business cases	All permits, classifications, levies, subsidies, and liabilities documented with status and timeline	Key regulatory elements listed; no jurisdiction analysis or timeline	Regulatory environment not mapped

Rule ID	Short Rule	Rationale	PRESENT criteria	PARTIAL criteria	ABSENT criteria
SYS-04	Design-maturity threshold before commitment	Build-before-design-freeze is the core overrun pattern	Open design points inventoried; design-freeze criteria defined; manufacturing/construction commitment conditional on maturity threshold	Design maturity acknowledged; no formal threshold or open-point register	Construction or supply-chain commitment pre-dates design freeze
SYS-05	Interface ownership and evidence	Unmanaged interfaces become the critical path late	Interface register exists; every critical interface has a named owner and a defined maturity evidence requirement	Key interfaces listed; ownership partial or informal; no maturity evidence	Interfaces not identified; no register
SYS-06	Risk-adjusted estimate alongside point estimate	Point budgets create false certainty	P50 and P80 or P90 estimates with Monte Carlo or equivalent; contingency sized to spread, not percentage	Range estimates produced but without simulation or explicit drivers	Single-point estimate only
SYS-07	Outside view before sponsor view	Internal teams systematically underestimate	Reference class applied; optimism-bias uplift calculated; estimate compared to comparable projects	Reference class mentioned; no calibration or uplift applied	No outside view; estimate based on internal assumptions only
SYS-08	Integration modelled as its own megaproject	Integration planned too late and too small	Integration and test phase has its own plan, budget, team, and operational-readiness evidence steps	Integration acknowledged; plan exists but is underweighted relative to civil/engineering spend	Integration phase not separately planned

Rule ID	Short Rule	Rationale	PRESENT criteria	PARTIAL criteria	ABSENT criteria
SYS-09	Supply-chain and workforce readiness	Immature supply chain is a system risk, not a procurement risk	Top-10 suppliers or critical trades assessed for single-point-of-failure; QA maturity and capacity confirmed	Key suppliers identified; no formal readiness assessment	Supply-chain risk treated as procurement task only
SYS-10	Stakeholders as system components	Social licence failure tips projects into legal or political paths	Stakeholder map includes who can delay, how, and from when; engagement strategy exists with legitimacy assessment	Stakeholder map sketched; no consequence modelling	Stakeholders listed as communications targets only
SYS-11	Business case stress-tested under policy and price scenarios	Fragile economics do not survive real market paths	Scenario matrix covers commodity price, demand, policy, and rate shocks; business case viable at P20 conditions	Sensitivity analysis performed; no combined or downside scenario	Business case modelled at single base case
SYS-12	Kill criteria explicitly defined	Without exit criteria, sunk costs drive continuation	Three or more specific findings that trigger scope change, resequencing, or project termination are documented	Kill concept acknowledged; criteria vague or aspirational	No kill criteria; project defined as unconditional commitment
SYS-13	Operations and maintenance in scope of definition	ORAT and handover gaps discovered only at commissioning	Operating model, O&M cost basis, data hand-over requirements, and ORAT steps are within the project definition scope	O&M mentioned; no operating model or ORAT plan	O&M treated as a post-FID problem

Rule ID	Short Rule	Rationale	PRESENT criteria	PARTIAL criteria	ABSENT criteria
SYS-14	FOAK treated as learning programme	First-of-a-kind projects mismanaged as if they were serial delivery	FOAK-specific assumptions documented; reserves sized for learning curve; knowledge-capture programme defined	FOAK risk acknowledged; reserves not differentiated from NOAK contingency	FOAK project scoped, costed, and contracted as if it were a repeat project
SYS-15	Cluster and platform questions escalated to portfolio level	Single-project optimisation produces sub-optimal network solutions	Project is explicitly positioned within a platform or cluster strategy; network effects modelled	Portfolio context acknowledged; no formal platform analysis	Project treated as a standalone asset regardless of network dependencies

Appendix B — Omission Vocabulary

The twelve standardised flags are used verbatim in all audit outputs. Their purpose is to produce a machine-readable, comparable signal across projects and gate reviews:

Flag	Triggered when
Full value chain not modelled	SYS-01 ABSENT or PARTIAL
Regulatory due diligence missing	SYS-03 ABSENT
Interface owner unknown	SYS-05 ABSENT or PARTIAL (no named owner)
Integration proof absent	SYS-08 ABSENT
Outside view missing	SYS-07 ABSENT
Kill criteria undefined	SYS-12 ABSENT
Design maturity not validated	SYS-04 ABSENT
Supply-chain readiness untested	SYS-09 ABSENT
Business case not stress-tested	SYS-11 ABSENT
FOAK not treated as learning programme	SYS-14 ABSENT (FOAK projects only)
Stakeholder legitimacy not mapped	SYS-10 ABSENT

Flag	Triggered when
O&M not in scope of definition	SYS-13 ABSENT

Appendix C — Evaluation Benchmark Detail

Eval	Configuration	Pass rate	Passes / Total	Time (s)	Tokens
Offshore wind + hydro-gen	With skill	100%	8/8	181.0	26,768
Offshore wind + hydro-gen	Without skill	38%	3/8	447.9	30,026
Hospital EHR transformation	With skill	100%	7/7	178.7	26,610
Hospital EHR transformation	Without skill	43%	3/7	267.9	21,559
FOAK lithium refinery	With skill	100%	7/7	212.2	27,038
FOAK lithium refinery	Without skill	43%	3/7	289.4	22,172
Aggregate with skill		100%	22/22	190.6	26,805
Aggregate without skill		41%	9/22	335.1	24,586

References

1. Flyvbjerg, B. & Gardner, D. (2023). *How Big Things Get Done*. Macmillan. Database of 16,000 projects; 8.5% delivered on cost and schedule; 0.5% delivered all promised benefits.
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